Topic 6

Frequency Response of FIR Filters

- Sinusoidal response of FIR systems
- Magnitude and phase response
- Delay, first difference, cascades, moving average
- Filtering sampled continuous-time signals

$$H(e^{j\hat{\omega}})$$

Key idea. Frequency response tells how an FIR filter changes each sinusoidal component.

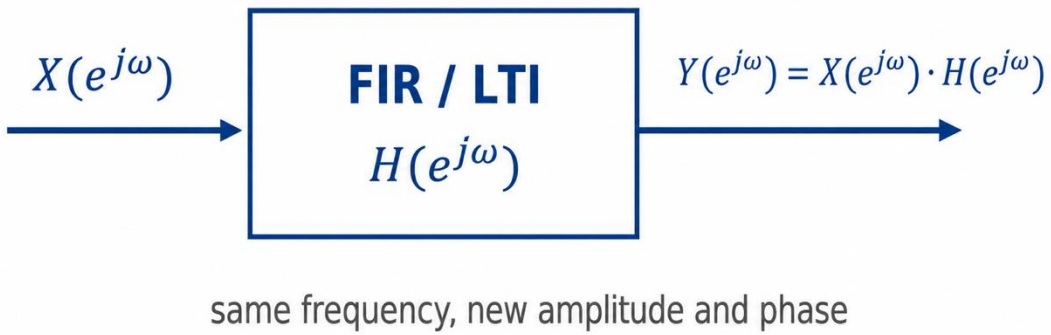
Sinusoidal response of FIR systems

Complex exponentials are eigenfunctions of LTI filters

$$y[n] = \sum_{k=0}^{M} b[k]\, x[n-k]$$

$$x[n] = X\,(e^{j\hat{\omega}n})$$

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^{M} b[k]\, e^{-j\hat{\omega}k}$$



- The frequency does not change.
- The complex gain H changes amplitude and phase.

Key idea. For LTI systems, sinusoids pass through as sinusoids at the same frequency.

From $H(e^{j\hat{\omega}})$ to real sinusoidal output

Magnitude scales; phase shifts

$$H(e^{j\hat{\omega}}) = |H(e^{j\hat{\omega}})|\, e^{j\angle H(ej\hat{\omega})}$$

$$A\cos(\hat{\omega}n+\varphi) \rightarrow A|H(e^{j\hat{\omega}})|\cos(\hat{\omega}n+\varphi+\angle H(e^{j\hat{\omega}}))$$

- The output amplitude is multiplied by the magnitude response.
- The output phase is shifted by the phase response.
- A zero in H cancels that sinusoidal component.

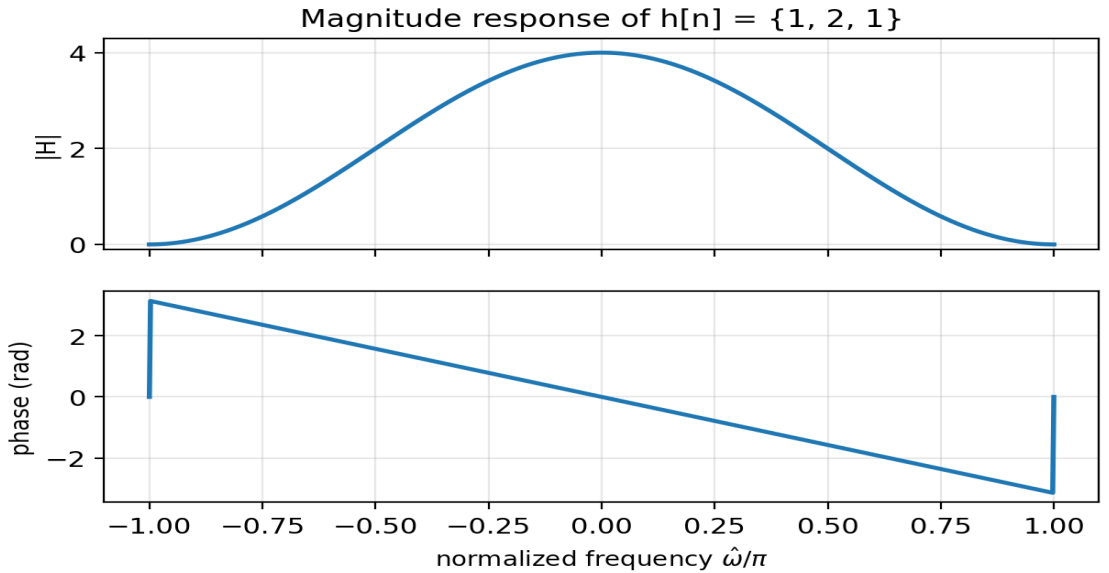
Key idea. The frequency response is a lookup table for every sinusoidal component.

Example 1: sinusoidal response

Find H and the steady-state output

$$y[n] = x[n] + 0.5x[n-1], \quad x[n]=4\cos(\pi n/2)$$

1. $H(e^{j\hat{\omega}}) = 1 + 0.5e^{-j\hat{\omega}}$.
2. At $\hat{\omega}=\pi/2$: $H = 1 - 0.5j = 1.118e^{-j0.464}$.
3. Output amplitude: $4 \cdot 1.118 = 4.472$.
4. Final output: $y[n] = 4.472\cos(\pi n/2 - 0.464)$.



Key idea. Evaluate H at the input frequency; then scale and shift the cosine.

Superposition and frequency response

A sum of sinusoids is filtered component by component

$$x[n] = \sum_{r} X_r \, e^{j\hat{\omega}_r n}$$

$$y[n] = \sum_{r} X_r \, H(e^{j\hat{\omega}_r}) \, e^{j\hat{\omega}_r n}$$

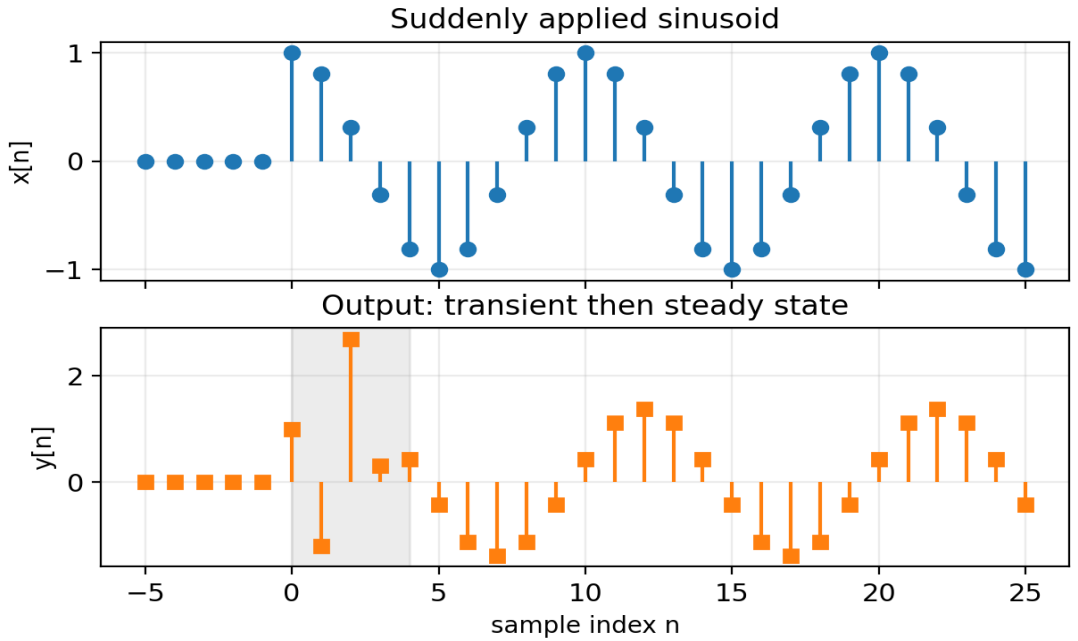
- Linearity lets us filter each component separately.
- No new frequencies are created by an LTI FIR filter.
- Components can be amplified, delayed, attenuated, or cancelled.

Key idea. Frequency response maps an input spectrum into an output spectrum.

Steady-state and transient response

A finite-onset sinusoid is not an eternal eigenfunction

- Before the input begins, the output is zero.
- During the first M samples, the FIR memory is only partly filled.
- After the transient, the sinusoidal steady-state formula applies.
- When the input ends, a second transient occurs.



Key idea. The sinusoidal-response rule describes the steady state, not the onset transient.

Properties of the frequency response

Periodicity and symmetry

$$H(e^{j(\hat{\omega}+2\pi)}) = H(e^{j\hat{\omega}})$$

$$\text{real } h[n] \Rightarrow H(e^{-j\hat{\omega}}) = H^*(e^{j\hat{\omega}})$$

- Discrete-time frequency response is periodic with period 2π.
- For real coefficients, magnitude is even.
- For real coefficients, phase is odd, except for wrapping and zeros.
- Usually one interval, such as $-\pi \le \hat{\omega} \le \pi$, is enough.

Key idea. The whole response is determined by one principal frequency interval.

Delay system

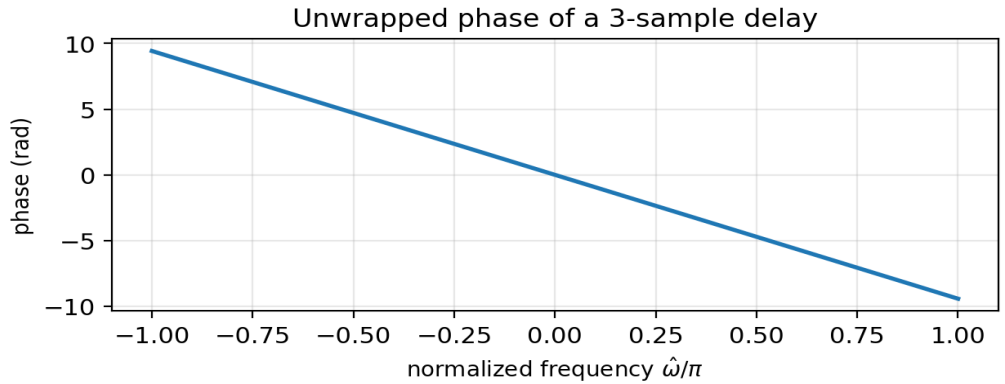
A pure delay changes phase but not magnitude

$$y[n] = x[n - n_0]$$

$$h[n] = \delta[n - n_0]$$

$$H(e^{j\hat{\omega}}) = e^{-j\hat{\omega}n_0}$$

- Magnitude: $|H| = 1$ for all frequencies.
- Phase: $\angle H = -\hat{\omega}n_0$ before wrapping.
- The slope of phase corresponds to delay.



Key idea. A pure delay preserves amplitude while imposing a linear phase shift.

Example 2: pure delay

Compute the output of a delayed sinusoid

$$y[n]=x[n-3], \quad x[n]=2\cos(0.4\pi n+\pi/6)$$

1. $H(e^{j\hat{\omega}}) = e^{-j3\hat{\omega}}$.
2. At $\hat{\omega}=0.4\pi$: phase shift $= -3(0.4\pi)=-1.2\pi$.
3. The amplitude is unchanged: $|H|=1$.
4. $y[n]=2\cos(0.4\pi n+\pi/6-1.2\pi)$.

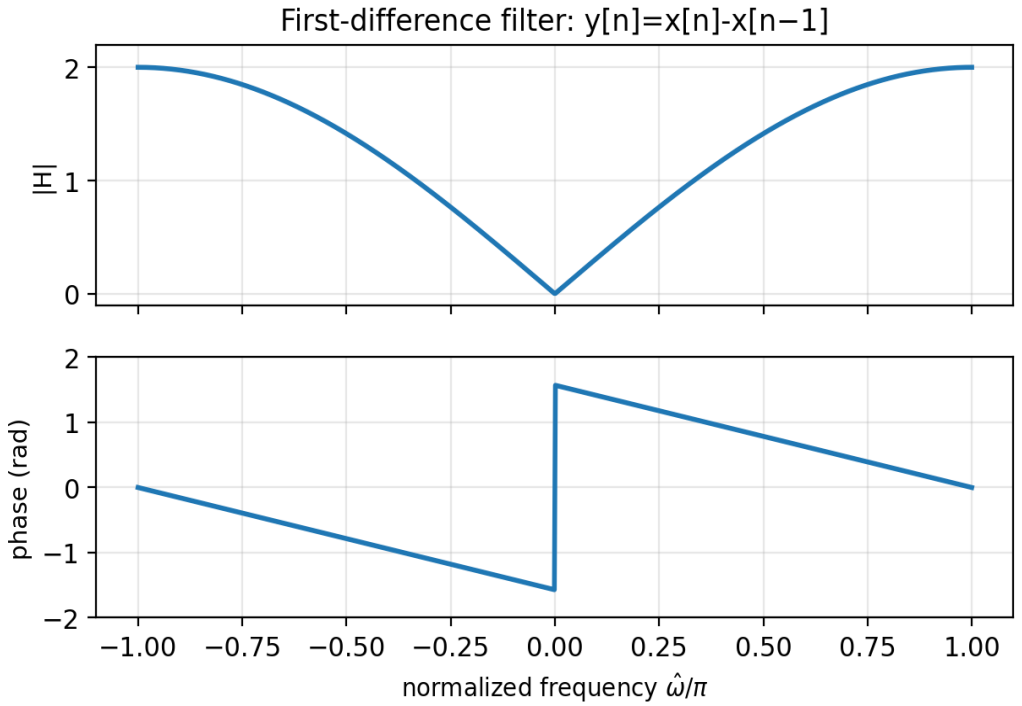
First-difference system

A simple high-pass FIR filter

$$y[n] = x[n] - x[n-1]$$

$$H(e^{j\hat{\omega}}) = 1 - e^{-j\hat{\omega}}$$

$$|H(e^{j\hat{\omega}})| = 2|\sin(\hat{\omega}/2)|$$



Key idea. The first difference removes DC and emphasizes rapid changes.

Example 3: first difference

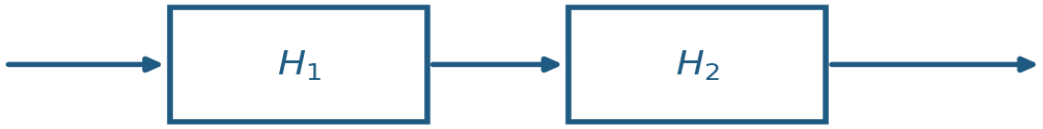
Evaluate gains and interpret

$$H(e^{j\hat{\omega}}) = 1 - e^{-j\hat{\omega}}$$

1. At DC: $H(e^{j0}) = 0$, so constant signals are removed.
2. At $\hat{\omega}=\pi/2$: $|H| = 2\sin(\pi/4)=\sqrt{2}$.
3. At $\hat{\omega}=\pi$: $|H| = 2\sin(\pi/2)=2$.
4. Gain increases with frequency ⇒ high-pass behaviour.

Cascaded LTI systems

Frequency responses multiply; impulse responses convolve



$$H_{eq}(e^{j\hat{\omega}}) = H_1(e^{j\hat{\omega}})H_2(e^{j\hat{\omega}})$$

$$h_{eq}[n] = h_1[n] * h_2[n]$$

- Cascade order does not matter for LTI systems.
- Zeros of either subsystem become zeros of the cascade.
- In time domain, the equivalent impulse response is $h_1 * h_2$.

Key idea. Cascades are easier to analyze in the frequency domain because multiplication replaces convolution.

Example 4: cascade

Two moving sums in sequence

$$h_1[n]=\{1,1\}, \quad h_2[n]=\{1,1\}$$

1. Frequency responses: $H_1=1+e^{-j\hat{\omega}}$, $H_2=1+e^{-j\hat{\omega}}$.
2. Overall response: $Heq=(1+e^{-j\hat{\omega}})^2$.
3. Equivalent impulse response: $heq=h_1*h_2=\{1,2,1\}$.
4. The cascade is a three-point triangular smoother.

Key idea. Cascading two simple averagers produces a stronger smoother.

Moving-average filtering

A low-pass smoother with structured zeros

$$y[n] = (1/L) \sum_{k=0}^{L-1} x[n-k]$$

$$h[n] = 1/L, \quad 0 \leq n \leq L-1$$

- A longer average smooths more strongly.
- The main lobe becomes narrower.
- The phase corresponds to a delay of (L−1)/2 samples.
- Zeros cancel specific sinusoidal components.

Key idea. A moving average is simple in time domain but very informative in frequency domain.

Moving-average frequency response

Finite geometric series

$$H(e^{j\hat{\omega}}) = (1/L) \sum_{k=0}^{L-1} e^{-j\hat{\omega}k}$$

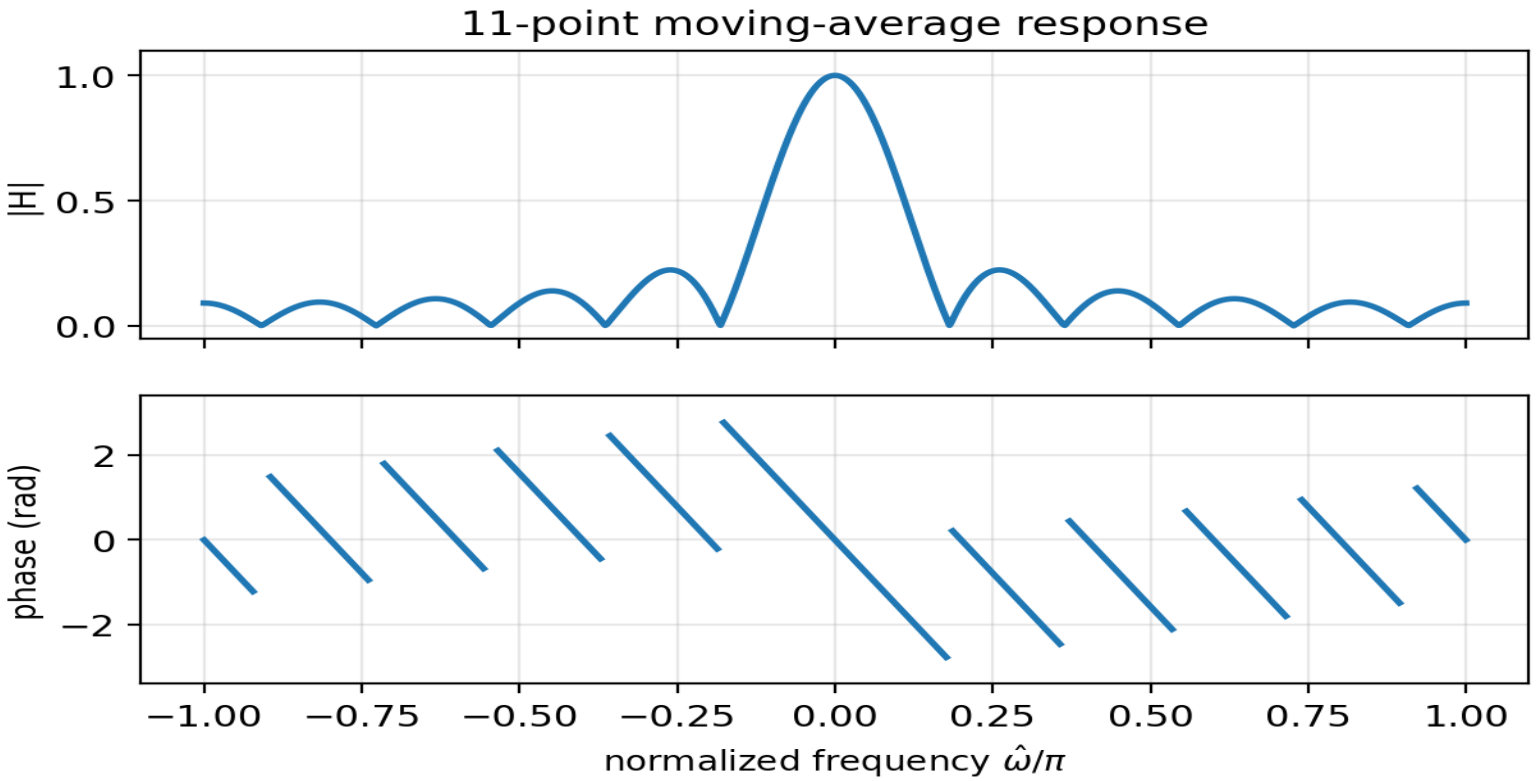
$$H(e^{j\hat{\omega}}) = (1/L) \cdot [\sin(L\hat{\omega}/2) / \sin(\hat{\omega}/2)] \cdot e^{-j\hat{\omega}(L-1)/2}$$

- Amplitude factor: Dirichlet / sinc-like shape.
- Phase factor: linear delay.
- Zeros occur when sin(Lω̂/2)=0, excluding DC.

Key idea. The closed form separates smoothing strength from delay.

11-point moving-average response

Magnitude response shows low-pass smoothing



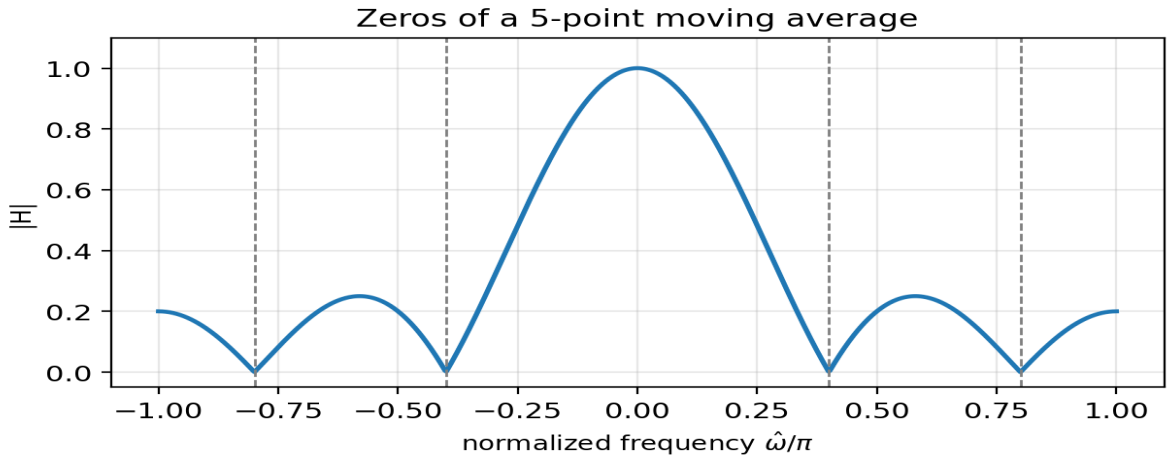
Key idea. Low frequencies near DC pass; many higher frequencies are attenuated or cancelled.

Example 5: moving-average zeros

Cancel a sinusoidal disturbance

Use an L-point moving average to cancel $\hat{\omega} = \pi/2$.

1. Zeros satisfy $\hat{\omega} = 2\pi m/L$, with $m=1,\ldots,L-1$.
2. Set $\pi/2 = 2\pi m/L \Rightarrow L = 4m$.
3. Smallest choice: $m=1$, so $L=4$.
4. Difference equation: $y[n]=(1/4)(x[n]+x[n-1]+x[n-2]+x[n-3])$.



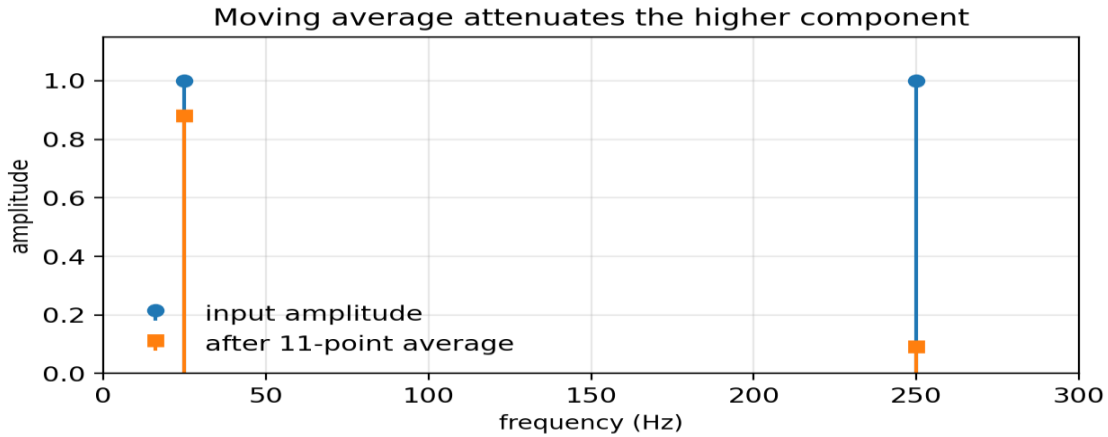
Filtering sampled continuous-time signals

Relating analogue and normalized frequency

$$\hat{\omega} = \omega T_s = 2\pi f / f_s$$

output gain at analogue frequency f: $|H(e^{j2\pi f/f_s})|$

- Sampling maps analogue frequency into normalized frequency.
- The discrete-time filter acts on that normalized frequency.
- After ideal reconstruction, the analogue frequency is preserved if there is no aliasing.



Key idea. Discrete-time frequency response predicts analogue behaviour after sampling and reconstruction.

Example 6: sampled signal

11-point average at fs = 1000 Hz

$$x(t)=\cos(2\pi\cdot 25t)+\cos(2\pi\cdot 250t)$$

1. Normalized frequencies: $\hat{\omega}_1=2\pi\cdot 25/1000=0.05\pi;\ \ \hat{\omega}_2=0.5\pi.$
2. Evaluate the 11-point moving-average response at both frequencies.
3. The 25-Hz component lies near DC and is mostly preserved.
4. The 250-Hz component lies in a low-gain region and is strongly attenuated.

Even- and odd-length moving averages

Delay can be integer or fractional

linear phase factor: $e^{-j\hat{\omega}(L-1)/2}$

delay $= (L-1)/2$ samples

- Odd L gives an integer-sample delay, e.g., L=5 → delay 2.
- Even L gives a half-sample delay, e.g., L=4 → delay 1.5.
- A half-sample delay is meaningful after continuous-time reconstruction.

Key idea. Linear phase preserves waveform shape apart from gain and delay.

Notation reference

Symbol	Meaning
$x[n]$, $y[n]$	Input and output discrete-time sequences
$h[n]$	Impulse response of the FIR filter
b_k	FIR filter coefficient at delay k
M	Filter order; the largest delay index
$L=M+1$	Filter length or number of coefficients
$H(e^{j\hat{\omega}})$	Frequency response evaluated at normalized radian frequency $\hat{\omega}$
$\lvert H(e^{j\hat{\omega}})\rvert$	Magnitude response or frequency-dependent gain
$\angle H(e^{j\hat{\omega}})$	Phase response
$\hat{\omega}$	Normalized radian frequency, in radians per sample
ω	Continuous-time radian frequency, in radians per second
T_s	Sampling period
$f_s=1/T_s$	Sampling rate
$*$	Convolution operator
$\delta[n]$	Unit impulse sequence
$u[n]$	Unit-step sequence

Key takeaways: FIR frequency response

1 One frequency in → the same frequency out

For an LTI filter, a sinusoid changes amplitude and phase, not frequency.

2 Read the frequency response

$|H(e^{j\hat{\omega}})|$ tells the gain; $\angle H(e^{j\hat{\omega}})$ tells the phase shift.

3 Predict the output of a sinusoid

Find $\hat{\omega}$, evaluate H at that frequency, then scale the amplitude and add the phase.

4 Understand the basic FIR examples

A delay changes phase; a first difference suppresses DC; a moving average smooths.

5 Combine filters and components

Filter each sinusoid separately (superposition); cascaded responses multiply.

6 Recognize zeros, transients, and sampling

$H = 0$ cancels a component; onset produces transients; $\hat{\omega} = 2\pi f/f_s$ connects Hz to digital frequency.

Practical rule: $y[n] = A|H(e^{j\hat{\omega}})| \cos(\hat{\omega}n + \phi + \angle H(e^{j\hat{\omega}}))$